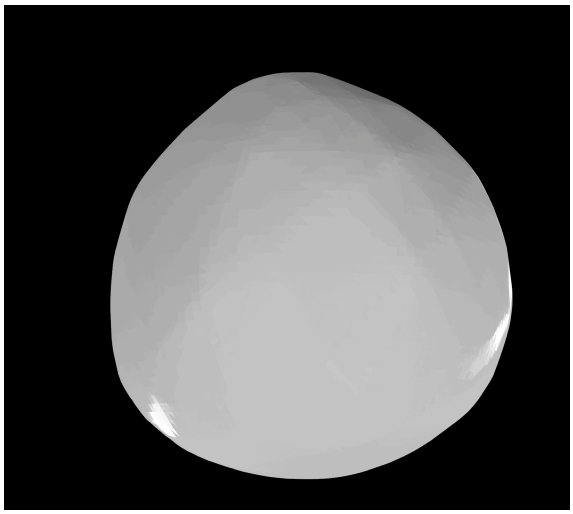
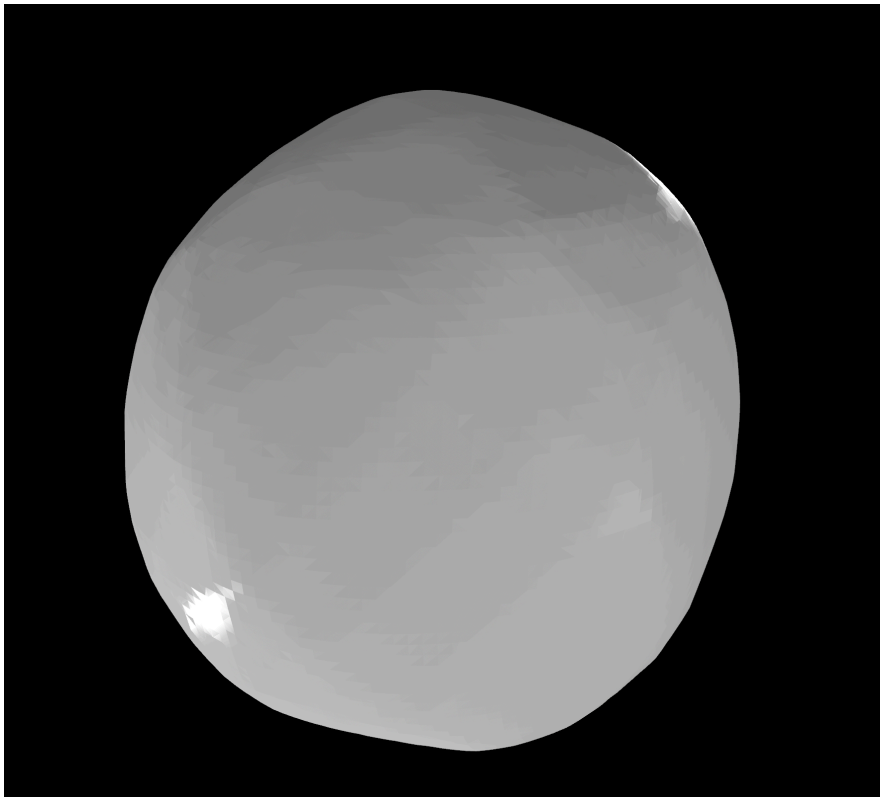
For my starter project, I implemented a pipeline for learning a neural implicit representation of a procedurally generated family of 3D shapes.

First, I generated a synthetic dataset based on the superellipsoid family of shapes. By randomly sampling parameters controlling axis scale and curvature, I created a diverse set of related 3D shapes. For each shape, I sampled many 3D points within a bounding cube and evaluated an implicit superellipsoid function to produce an approximate signed distance value for each point. This produced a dataset of point coordinates paired with scalar SDF values.

I then implemented a latent-conditioned neural implicit model in PyTorch. The network takes a 3D coordinate together with a learned latent vector representing a specific shape and predicts the SDF value at that point. Each shape in the dataset has its own latent embedding, and both the network parameters and latent vectors are optimized during training.

Once trained, the model defines a continuous scalar field over space. I extract surfaces from this field using marching cubes, allowing the learned representation to be converted into a mesh.

Results show that the network can reconstruct shapes from the dataset and that interpolation between latent vectors produces smooth transitions between shapes. This suggests that the model learns a continuous representation of the underlying procedural shape family.

Some of the shapes generated are shown below (superellipsoids can assume any shape from a deformed cube to an almost perfect sphere sphere (most of the time it will look like a deformed egg or a sphere).